

* $g: (\underbrace{t_1, \dots, t_s}_s, \underbrace{-1, \dots, -1}_t) \quad \text{pseudo-Riemannian metric is a section}$
 $g \in \Gamma(T^*M \otimes T^*M)$

which at each pt $x \in M$ defines a non-deg, symmetric bilinear

$$g_x: T_x M \times T_x M \rightarrow \mathbb{R}$$

of signature (s, t)

* The Frame Bundle has the structure of a principal $O(s, t)$ -Bundle

$$G \rightarrow P \xrightarrow{\pi} M, \quad G' \rightarrow P' \xrightarrow{\pi'} M'$$

$f: G \rightarrow G'$ be a group homomorphism

Bundle morphism (f -equivariant)

$$H: P \rightarrow P' \quad \text{if}$$

$$P \xrightarrow{H} P'$$

$$\begin{array}{ccc} & & \\ & \searrow \pi & \swarrow \pi' \\ & M & \end{array}$$

and

$$\boxed{\begin{aligned} \pi' \circ H &= \pi \\ H(p \cdot g) &= H(p) \cdot f(g) \end{aligned}}$$

$\forall p \in P, g \in G$
 f -reduction of P'

* If $f: G \hookrightarrow G'$ is an embedding, then H is called a G -reduction of P' and the image of H is a principal G -subbundle.

Def: (a) (M, g) is orientable if the frame bundle can be reduced to a principal $SO(s, t)$ Bundle under the embedding $SO(s, t) \subset O(s, t)$

(b) (M, g) is time-orientable if the frame bundle can be reduced to a principal $O^+(s, t)$ -bundle under the embedding $O^+(s, t) \subset O(s, t)$

(c) (M, g) (a) $\oplus$ (b) if $SO^+(s, t) \subset O(s, t)$

$\Rightarrow$ Time orientability $\Rightarrow A_0^0 \geq 1$

$$g(\Delta x, \Delta x) = g(x, x)$$

$$\Delta^T g \Delta = g$$

$$\Delta_\mu^\sigma g_{\sigma\lambda} \Delta_\nu^\lambda = g_{\mu\nu}$$

$$-1 = \Delta_0^0 g_{00} \Delta_0^0 + \Delta_0^i g_{ij} \Delta_0^i$$

$$\Rightarrow 1 = (\Delta_0^0)^2 - (\Delta_0^i)^2$$

$$\Rightarrow |\Delta_0^0| \geq 1$$

$L \rightarrow \Lambda^0 \geq 1$ (time preserving)

* Using Homotopy + Grassmanns: TM admits maximally g-pos def. vec. subbundles vec. subbundles $W \rightarrow M$ and that any two such bundles and any two such are homotopic

$\rightarrow$ A time orientation: Corresponds to a choice of orientation of such a maximally g-pos. def. vec. subbundle W .

* Let $\lambda: \text{SO}(n) \rightarrow \text{GL}(V)$ be a rep. of $\text{SO}(n)$ on $\mathbb{K}$ -vec space V . The associated vec. Bundle of $P^{\text{SO}}(M)$ is defined

$$P^{\text{SO}}(M) \times_{\lambda} V := P^{\text{SO}}(M) \times V / \sim$$

$$(h_1, v_1) \sim (h_2, v_2) : \Leftrightarrow \exists A \in \text{SO}(n) : h_2 = h_1 \circ A, v_2 = \lambda(A^{-1})v_1$$

• Denote the equivalence class of (h, v) as

$$[h, v] \in P^{\text{SO}} \times_{\lambda} V$$

$\rightarrow$ We are giving V onto the fibers of the frame Bundle

• Induced projection $\tilde{\pi} : P^{\text{SO}}(M) \times_{\lambda} V \rightarrow M$

$$\tilde{\pi}([h, v]) := \pi(h)$$

$$A \in \text{SO}(n)$$

Well defined since $(\tilde{h}, \tilde{v}) \sim (h, v) \Rightarrow \tilde{h} = h \circ A, \tilde{v} = \lambda(A^{-1})v$

$$\tilde{\pi}([\tilde{h}, \tilde{v}]) = \tilde{\pi}([h \circ A, \lambda(A^{-1})v]) = \pi(h \circ A) = \pi(h)$$

• Vector space structure:

$$\alpha[h, v_1] + \beta[h, v_2] := [h, \alpha v_1 + \beta v_2]$$

• Standard matrix rep $\lambda: \text{SO}(n) \hookrightarrow \text{GL}(n, \mathbb{R})$

$$P^{\text{SO}}(M) \times_{\lambda} \mathbb{R}^n \xrightarrow{\cong} TM$$

$$[h, v] \mapsto h(v)$$

Def: Let M be an oriented Riemann Mfd. A spin structure on M is a pair $(P^{\text{Spin}}(M), \tilde{\lambda})$ consisting of

a) A spin-principal bundle $P^{\text{Spin}}(M)$ over M s.t.

b) a twofold covering: $\tilde{\lambda}: P^{\text{Spin}}(M) \rightarrow P^{\text{SO}}(M)$ s.t.

$$\begin{array}{ccc} P^{\text{Spin}}(M) \times \text{Spin}(n) & \xrightarrow{\quad} & P^{\text{Spin}}(M) \xrightarrow{\pi_{\text{Spin}}} M \\ \tilde{\text{Ad}} \times \lambda \downarrow & & \downarrow \tilde{\text{Ad}} \\ P^{\text{SO}}(M) \times \text{SO}(n) & \xrightarrow{\quad} & P^{\text{SO}}(M) \xrightarrow{\pi_{\text{SO}}} M \end{array}$$

commutes. where $\tilde{\text{Ad}}: \text{Spin}(n) \rightarrow \text{SO}(n)$ is the double cover

→ core right actions

$$\lambda = \text{Ad}$$

* Spin structure is a Ad -equiv. bundle
or a λ -reduction of $\text{so}^+(n)$

morphism $\tilde{\text{Ad}}: \text{Spin}^+(n) \rightarrow \text{so}^+(n)$

$\tilde{\text{Ad}}$: twisted Adjoint

Recall $\text{Ad}: \text{Spin}(n) \rightarrow \text{SO}(n)$

$$\tilde{\text{Ad}}(a): V \rightarrow V$$

Short exact sequence:

$$1 \rightarrow A \xrightarrow{i} B \xrightarrow{p} C \rightarrow 1$$

Exact, $\text{Ker}(p) = \text{Im}(i)$

• This implies: $i: A \rightarrow B$ is injective

$\quad \quad \quad p: B \rightarrow C$ is surjective

Claim: $1 \rightarrow \mathbb{Z}_2 \rightarrow \text{Spin}(n) \xrightarrow{\tilde{\text{Ad}}} \text{SO}(n) \rightarrow 1$

(a) The map $\tilde{\text{Ad}}: \text{Spin}(n) \rightarrow \text{SO}(n)$ is surjective

(b) $\text{Ker}(\tilde{\text{Ad}}) = \mathbb{Z}_2 = \{-1, 1\}$

$$-1 = e_1 \cdot e_1 \in \text{Spin}(n)$$

$$\text{Ad}(-1)(x) = (-1)x(-1)^{-1} = x$$

$$\text{Ker}(\text{Ad}) = \{a \in \text{Spin}(n) \mid \text{Ad}(a) = \text{id}\}$$

$$\text{Ad}(1)(x) = x$$

$$\text{Thus } \{-1, 1\} \subset \text{Ker}(\text{Ad})$$

• let $a \in \text{Ker}(\text{Ad})$, then

$$x = \text{Ad}(a)(x) = axa^{-1}$$

$$ax = xa$$

$$a \in \mathbb{Z}(\text{Cl}_n^o) \text{ center of } \text{Cl}_n, \quad a \in \text{Spin}(n) \subset \text{Cl}_n^o$$

$$\mathbb{Z}(\text{Cl}_n) \cap \text{Cl}_n^o = \mathbb{R} \cdot 1$$

$$a = a \cdot 1 = v_1 \dots v_m, \quad v_i \in S^{n-1}$$

$$|a| = |v_1 \dots v_m| = 1 \Rightarrow a = \pm 1 \quad \checkmark \quad \square$$

* In general $\lambda = \text{Ad}$